Number Theory Tutorial

Divisibility, GCD, Modular Arithmetic and CRT

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September 24, 2026

1. Divisibility

Recall that $a \mid b$ means that there exists an integer k such that

$$b = ak.$$

1. Prove that if $a \mid b$ and $a \mid c$, then
$$a \mid (b+c), \qquad a \mid (b-c).$$
2. Prove that if $a \mid b$ and $b \mid c$, then $a \mid c$.
3. Show that if $a \mid b$, then $a \mid kb$ for every $k \in \mathbb{Z}$.

2. Division Algorithm and GCD

For integers a and $b > 0$, there exist unique integers q, r such that

$$a = bq + r, \qquad 0 \le r < b.$$

The greatest common divisor of a and b is denoted by

$$\gcd(a, b).$$

1. Use the Euclidean algorithm to compute:
$$\gcd(252, 105).$$
2. Use the Euclidean algorithm to compute:
$$\gcd(1071, 462).$$
3. Find integers x, y such that
$$\gcd(56, 15) = 56x + 15y.$$
4. Prove that
$$\gcd(a, b) = \gcd(a, b - ka)$$
for every $k \in \mathbb{Z}$.
5. Find all positive integers n satisfying
$$\gcd(n, 18) = 6.$$

3. LCM and the Fundamental Theorem of Arithmetic

The least common multiple of a and b is denoted by

$$\operatorname{lcm}(a, b).$$

Every integer $n > 1$ can be written uniquely, up to the order of the factors, as a product of primes.

1. Prove that if p is prime and $p \mid ab$, then
$$p \mid a \quad \text{or} \quad p \mid b.$$
2. Verify that
$$\gcd(a, b) \operatorname{lcm}(a, b) = ab.$$
3. How many positive divisors does
$$2^x \cdot 3^y \cdot 5$$
have?

4. Modular Arithmetic

We write

$$a \equiv b \pmod{n}$$

if $n \mid (a-b)$.

1. Find the remainder of
$$2^{10}$$
modulo 7.
2. Find the remainder of
$$3^{100}$$
modulo 7.
3. Find the last digit of
$$7^{2026}.$$
4. Show that if
$$a \equiv b \pmod{n},$$
then
$$a^k \equiv b^k \pmod{n}$$
for every positive integer k.

5. Chinese Remainder Theorem

The Chinese Remainder Theorem says that if

$$\gcd(m,n) = 1,$$

then the system

$$x \equiv a \pmod{m}, \qquad x \equiv b \pmod{n}$$

has a unique solution modulo mn.

1. Solve
$$x \equiv 2 \pmod{3}, \qquad x \equiv 1 \pmod{5}.$$
2. Solve
$$x \equiv 1 \pmod{4}, \qquad x \equiv 3 \pmod{7}.$$
3. Solve
$$x \equiv 2 \pmod{5}, \qquad x \equiv 4 \pmod{9}.$$
4. Find the smallest positive integer satisfying
$$x \equiv 3 \pmod{7}, \qquad x \equiv 5 \pmod{11}.$$
5. Solve the system
$$x \equiv 1 \pmod{3}, \qquad x \equiv 2 \pmod{5}, \qquad x \equiv 4 \pmod{7}.$$
6. Find all solutions modulo 60 of
$$x \equiv 1 \pmod{3}, \qquad x \equiv 2 \pmod{4}, \qquad x \equiv 3 \pmod{5}.$$

6. Mixed Problems

1. Find
$$\gcd(2026, 782)$$
using the Euclidean algorithm.
2. Find the last two digits of
$$7^{100}.$$

3. Determine all integers x satisfying

$$x \equiv 2 \pmod 5, \qquad x \equiv 3 \pmod 7.$$

4. Find the smallest positive integer n such that

$$\gcd(n, 60) = 12$$

and

$$n \equiv 3 \pmod 7.$$

5. Prove that for every integer n,

$$n^3 - n$$

is divisible by 6.

6. Find the remainder when

$$2^{100} + 3^{100}$$

is divided by 7.

7. Find all positive integers n such that

$$\operatorname{lcm}(n, 12) = 60.$$

8. Solve

$$4x \equiv 6 \pmod{14}.$$

9. Find the smallest positive integer satisfying

$$x \equiv 1 \pmod 4, \qquad x \equiv 2 \pmod 5, \qquad x \equiv 3 \pmod 7.$$

10. Let p be a prime. Prove that if

$$p \mid a^2,$$

then

$$p \mid a.$$